ADVANCED PROBLEMS FOR LECTURE 15

In the lecture, we will be talking at a more general level of the patterns of what is going on. But there are deeper questions to explore therein, which we cannot get to. This is your opportunity to investigate deeper for yourself. Don't limit yourself to just these questions, let your curiosity guide you too! That is, if you wonder about any details, or if any certain patterns or ideas exist, see if you can state your question clearly, and answer it! Here are a few suggestions to get you started.

The discussion of anti-derivative hides within it the notion of *differential equations*. For example, if we know that the derivative of some function f is equal to g, then we can write this as

$$f' = g \quad \text{or} \quad \frac{df}{dx} = g.$$

Finding an f that works here is tantamount to solving the above "differential equation".
We will explore some of these ideas further here.

(1) Show that if f and g are two functions on an interval (a, b) such that $f' = g'$, then $f = g + C$ where C is a constant.

(2) Show that if n is a natural number, and

$$\frac{d^n f}{dx^n} = k$$

for some constant $k \neq 0$, then f is a polynomial of order n.

(3) Find solutions to the differential equation

$$\frac{df}{dx} = f$$

(4) Find solutions to the differential equation

$$\frac{df}{dx} = kf$$

where $k \neq 0$ is some constant.

(5) If you are familiar with linear algebra, you have heard of the notion of a linear map. That is, a map $L : V \to W$ between two vector spaces is *linear* if

$$L(av_1 + bv_2) = aLv_1 + bLv_2$$

for any scalars a and b, and vectors v_1 and v_2.

(a) Show that the differentiation operator D, which acts on the space of differentiable functions on (a, b) (so the vectors are differentiable functions) defined by

$$Df = \frac{df}{dx}$$

is linear.

(b) Show that scalar multiples of the identity map are linear. The identity map is the *do nothing* map, i.e.

$$I(f) = f$$

(c) Show that question (3) can be rewritten as trying to find those functions f such that

$$(D - I)f = 0.$$

(d) Challenge: From your observation above, we are then trying to compute the kernels of the operators $(D - I)$ and $(D - kI)$. Can you argue that these kernels are one dimensional? Therefore telling us that we have found the complete set of solutions to the differential equations in (2) and (3)?